Saransh Saini

AI Engineer — Deep Learning — Computer Vision — LLMs — RAGs — PyTorch

+91-8178703402 • saransh.saini.ai@gmail.com • linkedin.com/in/saranshsaini48 • github.com/Saransh482003

ABOUT ME

AI Engineer with hands-on experience building deep learning and LLM-powered systems across computer vision, multimodal AI, retrieval augmented generation, and production ML. Worked on object detection, monocular depth estimation, OCR, ASR, and GraphRAG pipelines using PyTorch, YOLO, Neo4j, MLflow, and GCP.

SKILLS & ABILITIES

Programming and Databases: Python, Java, JavaScript, MySQL, Cypher (Neo4j)
Deep Learning / ML: PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, Ultralytics YOLO
LLM / GenAI: RAG, GraphRAG, RAPTOR, LangChain, Vector Search, Agent-based systems, Ollama
ML Deployment: MLOps, GCP, MLFlow, Docker, FastAPI, DVC, Git

EXPERIENCE

DataAnnotation.tech — *Contract AI Engineer* **May 2026 – Present**
- Conducted quality evaluations on pre-released coding agents, on metrics like task achievability, code quality, model thoroughness.

Healthcare Technology Innovation Centre, IITM RP — *AI/ML & Full-Stack Intern* **Jun 2025 – Feb 2026**
- Implemented RaptorRAG & GraphRAG architectures with LLM based summarization techniques and Neo4j.
- Assisted in the Network Connectivity module of an Organ Monitoring Device, making the Arduino system capable of communicating with the internet and publishing results on an online dashboard.

Centre for Industrial Consultancy and Sponsored Research, IIT Madras — *Teaching Assistant* **Jan 2025 – May 2025**
- Taught Tools in Data Science course to 1500+ students, dealing with LLMs, Data Scraping & Analytics.
- Worked on bulk assignments and project evaluations, ensuring timely feedback and academic integrity.

NEXT IAS — *Web Development Intern* **Jan 2024 – Apr 2024**
- Boosted event registrations by 15% through improvements to the website's UI/UX design.
- Worked Designed and developed new web pages, enhancing the website's overall structure and visual appeal.

PROJECTS

RAGner: A Tri-Modal RAG Framework Go to Project **Sep 2026**
- Architected a production-ready, dual-database retrieval system integrating Qdrant (Vector) for RAPTOR based semantic search and Neo4j Cloud (Graph) for entity relationship mapping, bypassing traditional LLM context-window limitations.
- Deployed the end-to-end pipeline to production, containerizing the backend inference engine via FastAPI on Render, while serving the client-side interface via Next.js on Vercel.

Action-Oriented Indoor Navigation Assistance Go to Project **Mar – Apr 2026**
- Developed an indoor navigation pipeline for visually impaired individuals. Using YOLO models for object detection and monocular depth models for depth estimation.
- Worked specifically on the object detection module, and also integration of all modules into one end-to-end pipeline.

VascCoach: A RAPTOR-based RAG System for Vascular Aging Go to Project **Dec 2025**
- Developed a RAPTOR-based RAG pipeline on 70+ medical research papers and device manuals. Using Hierarchical tree summarization technique to create the schema for the RAPTOP retrieval.
- Achieved an accuracy of about 70% on 50 precise questions derived from the 70+ documents. Also, explored the GraphRAG approach, using Neo4j as the graph database.

Heart Disease Detection MLOps Pipeline Go to Project **Aug 2025**
- Developed a production-grade heart disease classification system on GCP, containerizing a Flask inference API with Docker and orchestrating deployment on a Kubernetes (GKE) cluster for scalable real-time serving.
- Integrated MLFlow for experiment tracking and Evidently AI for continuous monitoring of data drift and model decay, ensuring long-term reliability of predictions in a production environment.

EDUCATION

Indian Institute of Technology, Madras (IIT Madras) **Jan 2022 – May 2026**
Bachelor of Science (BS) in Data Science and Application (4-year program) — **8.75 CGPA**

ACHIEVEMENTS & PUBLICATIONS

- **Second Runner Up, BharatGen 2025 Hackathon:** Won Rs. 30,000 for **CureCue**, an offline-first AI medical assistant using voice and image interaction.
- **Technical Report (ResearchGate):** Published a multi-model pipeline integration of **YOLOv8** (92.3% mAP) and **Gemini 2.0 Flash** for automated patient self-care, with **Prof. Neelesh S Upadhye**, Dept. of Mathematics, IIT Madras.